

Mechanism and Machines

Unit II - Mechanical Engineering

ME10008

Comprehensive Study Notes

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1 Introduction to Theory of Machines

Definition

Theory of Machines is a subject which deals with the relative motion between parts of machines and the forces acting between them.

1.1 Sub-divisions of Theory of Machines

1.1.1 1. Kinematics

Definition

Deals with the relative motion between various parts of machines (without considering forces).

1.1.2 2. Dynamics

Definition

Study of forces and their effects while acting upon machine parts in motion.

1.1.3 3. Kinetics

Definition

Focuses on inertia forces arising from the combined effect of mass and motion of machine parts.

1.1.4 4. Statics

Definition

Study of forces and their effects while machine parts are at rest. Mass of parts is assumed negligible.

2 Types of Motion

2.1 1. Rectilinear Motion (Translatory Motion)

Definition

The simplest type of motion along a straight line path.

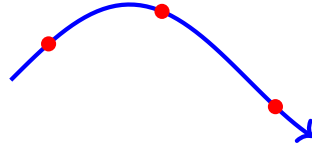


Straight Line Motion

2.2 2. Curvilinear Motion

Definition

Motion along a curved path. When confined to one plane, called plane curvilinear motion.



Curved Path Motion

2.3 3. Planar Motion

Definition

When motion of a body is confined to only one plane. May be rectilinear or curvilinear.

3 Kinematic Link or Element

Definition

Each part of a machine which moves relative to some other part is known as a kinematic link or element.

Important Note: A link may consist of several parts rigidly fastened together so they don't move relative to one another.

3.1 Types of Links

3.1.1 1. Rigid Link

Definition

A link which does not undergo any deformation while transmitting motion.

Note: Strictly speaking, rigid links don't exist. However, links with negligible deformation (connecting rod, crank) are considered rigid.

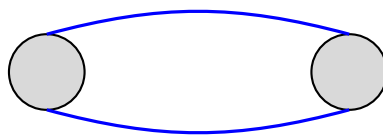
Examples: Connecting rod, crank of steam engine

3.1.2 2. Flexible Link

Definition

A link which is partly deformed but does not affect transmission of motion. Transmits tensile forces only.

Examples: Belts, ropes, chains, wires



Belt Drive (Flexible Link)

3.1.3 3. Fluid Link

Definition

A link formed by having fluid in a receptacle. Motion transmitted through fluid by pressure or compression only.

Examples: Hydraulic presses, jacks, brakes

4 Structure vs Machine

4.1 Structure

Definition

An assemblage of resistant bodies (members) having NO relative motion between them, meant for carrying loads.

Examples: Railway bridge, roof truss, machine frames

4.2 Difference Between Machine and Structure

Aspect	Machine	Structure
Motion	Parts move relative to one another	Members do NOT move relative to one another
Energy	Transforms energy into useful work	No energy transformation
Function	Transmits both power and motion	Transmits forces only

5 Kinematic Pair

Definition

Two links or elements in contact with each other form a pair. If relative motion is completely or successfully constrained in a definite direction, it's called a kinematic pair.

5.1 Types of Constrained Motions

5.1.1 1. Completely Constrained Motion

Definition

Motion between a pair is limited to a definite direction irrespective of the direction of force applied.

Example: Piston in cylinder - can only move in one direction

5.1.2 2. Incompletely Constrained Motion

Definition

Motion between a pair can take place in more than one direction.

Example: Shaft in a bearing - can rotate and slide

5.1.3 3. Successfully Constrained Motion

Definition

Constrained motion is not completed by itself but by some other external means.

Example: Foot step bearing - constrained by external force (weight)

6 Classification of Kinematic Pairs

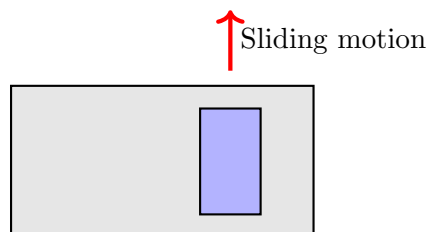
6.1 1. According to Type of Relative Motion

6.1.1 (a) Sliding Pair

Definition

Two elements connected such that one can only slide relative to the other.

Examples: Piston and cylinder, crosshead and guides, ram and guides in shaper



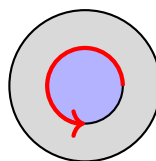
Piston-Cylinder (Sliding Pair)

6.1.2 (b) Turning Pair (Revolute Pair)

Definition

Two elements connected such that one can only turn or revolve about a fixed axis of another link.

Examples: Crankshaft in journal bearing, shaft with collars, cycle wheel on axle



Shaft in Bearing (Turning Pair)

6.1.3 (c) Rolling Pair

Definition

Two elements connected such that one rolls over another fixed link.

Examples: Ball bearings, roller bearings

6.1.4 (d) Screw Pair

Definition

Two elements connected such that one can turn about the other by screw threads.

Examples: Lead screw with nut, bolt with nut

6.1.5 (e) Spherical Pair

Definition

Two elements connected such that one (spherical shape) turns or swivels about the other fixed element.

Examples: Ball and socket joint, car mirror attachment, pen stand

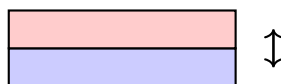
6.2 2. According to Type of Contact

6.2.1 (a) Lower Pair

Definition

Two elements have SURFACE contact when relative motion takes place. Surface of one slides over the other.

Examples: Sliding pairs, turning pairs, screw pairs



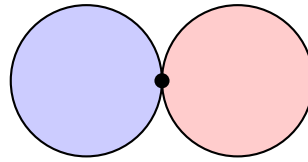
Surface Contact

6.2.2 (b) Higher Pair

Definition

Two elements have LINE or POINT contact when relative motion takes place. Motion is partly turning and partly sliding.

Examples: Friction discs, toothed gearing, roller bearings, cam and follower



Point Contact

6.3 3. According to Type of Closure

6.3.1 (a) Self-Closed Pair

Definition

Two elements connected mechanically such that only the required relative motion occurs.

Note: Lower pairs are self-closed pairs.

6.3.2 (b) Force-Closed Pair

Definition

Two elements NOT connected mechanically but kept in contact by external forces (spring, gravity).

Example: Cam and follower

7 Kinematic Chain

Definition

When kinematic pairs are coupled such that the last link joins the first link to transmit definite motion, it's called a kinematic chain.

7.1 Relations for Kinematic Chain

Relation 1: Between links (l) and pairs (p)

$$l = 2p - 4$$

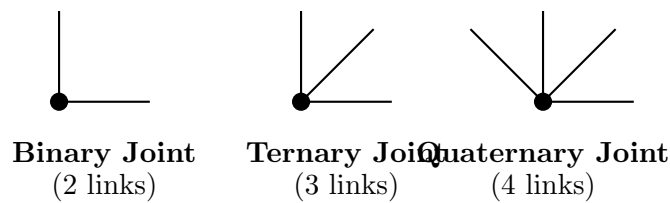
Relation 2: Between links (l) and joints (j)

$$j = \frac{3}{2}(l - 1)$$

or equivalently: $l = \frac{2j}{3} + 1$

For a valid kinematic chain: Both conditions must be satisfied. If satisfied, it forms a kinematic chain; otherwise, it's a locked chain (rigid frame).

7.2 Types of Joints in a Chain



8 Mechanism and Machine

8.1 Mechanism

Definition

When one link of a kinematic chain is fixed, the chain becomes a mechanism. Used for transmitting or transforming motion.

Classification:

- **Simple mechanism:** 4 links
- **Compound mechanism:** More than 4 links

Examples: Engine indicators, typewriter

8.2 Machine

Definition

When a mechanism is required to transmit power or do some particular work, it becomes a machine.

Key Point: Mechanism → Motion transmission — Machine → Power transmission + Work

9 Degrees of Freedom

Definition

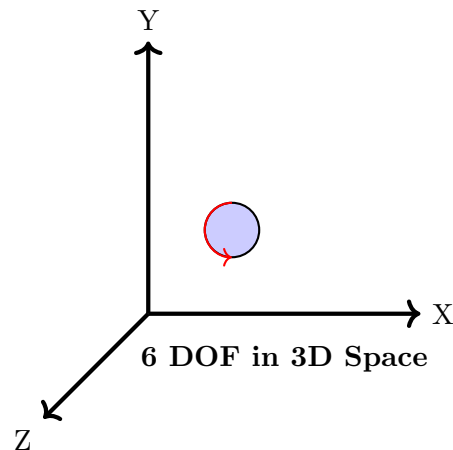
Number of input parameters (pair variables) that must be independently controlled to bring the mechanism into a useful engineering purpose.

Simple meaning: Number of ways a body can move in space.

9.1 Maximum Degrees of Freedom

For a link in 3D space: **6 degrees of freedom**

- 3 linear movements (along X, Y, Z axes)
- 3 rotational movements (about X, Y, Z axes)



9.2 Degrees of Freedom for Planar Mechanisms

Kutzbach Criterion (Grübler's Equation):

$$n = 3(l - 1) - 2j - h$$

where:

- n = degrees of freedom
- l = number of links
- j = number of binary joints (lower pairs)
- h = number of higher pairs

Special Cases:

- $n = 0$: Locked chain (structure)
- $n = 1$: Mechanism with definite motion
- $n = 2$: Mechanism requires 2 inputs
- $n < 0$: Redundant constraints

10 Inversion of Mechanism

Definition

Method of obtaining different mechanisms by fixing different links in turn in a kinematic chain.

Key Concept: One kinematic chain can produce multiple mechanisms by fixing different links.

11 Types of Kinematic Chains

The most important are those with 4 lower pairs (sliding or turning):

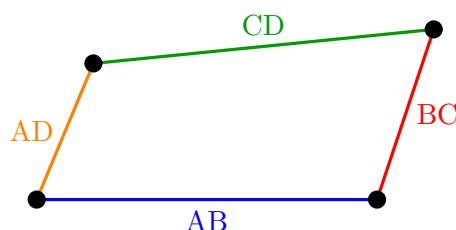
11.1 1. Four Bar Chain (Quadric Cycle Chain)

Grashof's Law

For continuous relative motion between two links, the sum of shortest and longest link lengths should not be greater than the sum of remaining two links.

$$L + S \leq P + Q$$

where: L = longest link, S = shortest link, P, Q = intermediate links



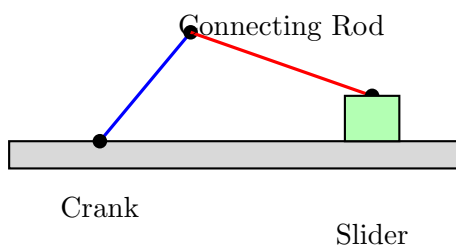
Four Bar Chain

11.1.1 Inversions of Four Bar Chain

1. **Beam Engine (Crank and Lever):** One crank, one lever
2. **Coupling Rod of Locomotive (Double Crank):** Two cranks
3. **Watt's Indicator (Double Lever):** Two levers

11.2 2. Single Slider Crank Chain

Consists of one sliding pair and three turning pairs.



Single Slider Crank Chain

11.2.1 Inversions of Single Slider Crank Chain

1. **Pendulum Pump (Bull Engine):** Crank fixed
2. **Oscillating Cylinder Engine:** Connecting rod fixed
3. **Rotary Internal Combustion Engine (Gnome Engine):** Cylinder fixed
4. **Crank and Slotted Lever Quick Return Mechanism:** Used in shapers

11.3 3. Double Slider Crank Chain

Contains two sliding pairs and two turning pairs.

12 Crank and Slotted Lever Mechanism

Used in shaping machines for quick return motion.

12.1 Important Parameters

Stroke Length:

$$L = \frac{2 \times AP \times CB}{AC}$$

where:

- AP = Length of slotted lever
- CB = Crank radius
- AC = Center distance

Time Ratio:

$$\text{Time ratio} = \frac{\beta}{\alpha} = \frac{360 - \alpha}{\alpha}$$

Percentage Times:

$$\begin{aligned}\text{Cutting time \%} &= \frac{\beta}{360} \times 100 \\ \text{Return time \%} &= \frac{\alpha}{360} \times 100\end{aligned}$$

12.2 Solved Example

Example

A crank and slotted lever mechanism has:

- Center distance $AC = 300$ mm
- Crank radius $CB = 120$ mm
- Slotted lever length $AP = 800$ mm

Find: (1) Stroke length, (2) Time ratio, (3) Percentage times

Solution:

(1) Stroke Length:

$$L = \frac{2 \times 800 \times 120}{300} = 640 \text{ mm}$$

(2) Finding angle α :

$$\begin{aligned}\sin\left(\frac{\alpha}{2}\right) &= \frac{CB}{AC} = \frac{120}{300} = 0.4 \\ \frac{\alpha}{2} &= 23.58 \\ \alpha &= 47.16 \approx 132.8\end{aligned}$$

(considering full angle)

$$\beta = 360 - 132.8 = 227.2$$

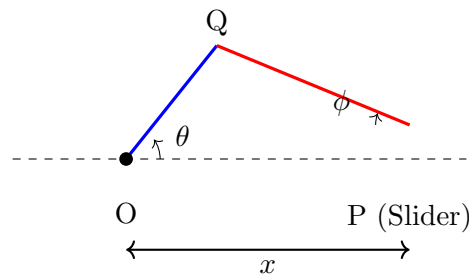
Time ratio: $= \frac{227.2}{132.8} = 1.71 : 1$

(3) Percentage Times:

$$\text{Cutting time \%} = \frac{227.2}{360} \times 100 = 63.1\%$$

$$\text{Return time \%} = \frac{132.8}{360} \times 100 = 36.9\%$$

13 Displacement in Slider Crank Mechanism



13.1 Key Formulas

Stroke Length:

$$L = 2r$$

where r = crank radius

Distance traveled per minute:

$$\text{Distance} = 2LN = 4rN$$

where N = RPM of crank

Relation between angles:

$$\frac{\sin \phi}{\sin \theta} = \frac{r}{l}$$

where l = connecting rod length, r = crank radius, $n = l/r$

$$\sin \phi = \frac{\sin \theta}{n}$$

Displacement of slider:

$$x = r \left[(1 - \cos \theta) + \frac{1}{n} (1 - \cos \phi) \right]$$

For small angles (approximate):

$$x \approx r \left[(1 - \cos \theta) + \frac{1}{2n} \sin^2 \theta \right]$$

13.2 Solved Example

Example

Given:

- Connecting rod length: $l = 160$ mm
- Ratio: $n = l/r = 4$
- Crank speed: $N = 6000$ RPM
- Crank angle: $\theta = 75$

Find: (1) Stroke length, (2) Distance per second, (3) Max angular displacement of rod, (4) Slider displacement at $\theta = 75$

Solution:

Crank radius: $r = \frac{l}{n} = \frac{160}{4} = 40$ mm

(1) Stroke Length:

$$L = 2r = 2 \times 40 = 80 \text{ mm}$$

(2) Distance per second: Distance = $\frac{2LN}{60} = \frac{2 \times 80 \times 6000}{60} = 16000$ mm/s = 16 m/s

(3) Max angular displacement of connecting rod:

Maximum occurs when $\theta = 90$

$$\sin \phi = \frac{\sin \theta}{n} = \frac{\sin 90}{4} = \frac{1}{4} = 0.25$$

$$\phi = \sin^{-1}(0.25) = 14.47$$

(4) Slider displacement at $\theta = 75$:

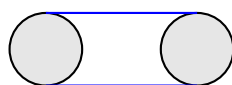
First find ϕ at $\theta = 75$: $\sin \phi = \frac{\sin 75}{4} = \frac{0.9659}{4} = 0.2415$ $\phi = 13.97$

$$x = r[(1 - \cos \theta) + \frac{1}{n}(1 - \cos \phi)] \quad x = 40[(1 - \cos 75) + \frac{1}{4}(1 - \cos 13.97)] \quad x = 40[(1 - 0.2588) + 0.25(1 - 0.9702)] \quad x = 40[0.7412 + 0.25 \times 0.0298] \quad x = 40[0.7412 + 0.00745] = 40 \times 0.7487 = 29.95 \text{ mm}$$

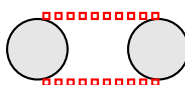
Approximately: $x \approx 30$ mm or using more precise calculation: $x = 34.31$ mm

14 Power Transmission Systems

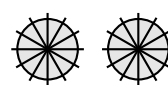
14.1 Three Main Types



Belt Drive



Chain Drive



Gear Drive

- **Belt Drive:** Flexible, smooth operation, slip possible
- **Chain Drive:** Positive drive, no slip, moderate distances
- **Gear Drive:** Positive drive, exact velocity ratio, short distances

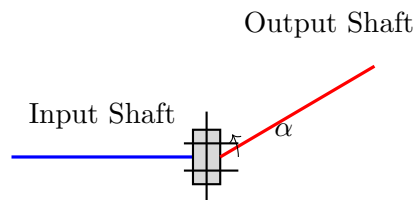
15 Hooke's Joint (Universal Joint)

Definition

A coupling used to connect two shafts whose axes are inclined to each other.

15.1 Applications

- Transmit power when driving and driven shafts are parallel but not coaxial
- Power transmission when shafts are not parallel and axes are intersecting
- Common in automobiles: engine to front/rear wheels in four-wheeled vehicles



Hooke's Joint

16 Gears

Definition

A toothed element used for transmitting rotary motion from one shaft to another. Provides positive and smooth drive with exact velocity ratio.

Advantages:

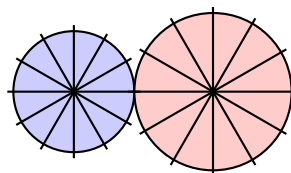
- Positive drive (no slip)
- Exact velocity ratio
- Compact design
- High efficiency

16.1 Classification of Gears

16.1.1 1. Based on Position of Shaft Axes

(a) Parallel Shafts:

- **Spur Gears:** Teeth parallel to shaft axis

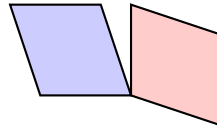


Spur Gears

- **Helical Gears:** Teeth at an angle to shaft axis, smoother operation
- **Double-Helical Gears (Herringbone):** Two helical gears with opposite helix angles

(b) Intersecting Shafts:

- **Straight Bevel Gears:** Teeth radial to point of intersection
- **Spiral Bevel Gears:** Teeth inclined, smoother engagement



Bevel Gears

(c) Non-parallel and Non-intersecting Shafts:

- **Skew Bevel Gears (Skew Spiral Gears)**
- **Worm and Worm Wheel**

16.1.2 2. Based on Peripheral Velocity

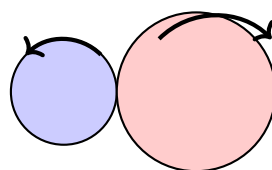
- **Low velocity:** $v < 3$ m/s
- **Medium velocity:** $3 < v < 15$ m/s
- **High velocity:** $v > 15$ m/s

16.1.3 3. Based on Position of Teeth

- **Straight:** Spur gears
- **Inclined:** Helical gears
- **Curved:** Spiral gears

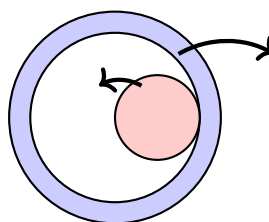
16.1.4 4. Based on Type of Gearing

(a) External Gearing:



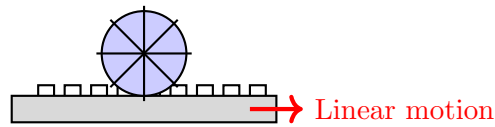
External Gears (Opposite rotation)

(b) Internal Gearing:



Internal Gears (Same rotation)

(c) Rack and Pinion:



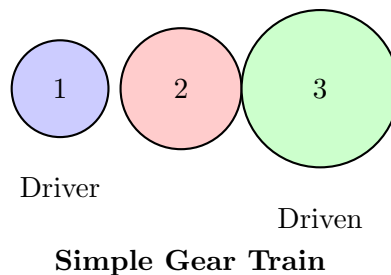
Rack and Pinion (Rotary to Linear)

17 Types of Gear Trains

17.1 1. Simple Gear Train

Definition

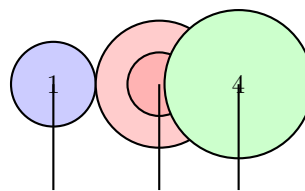
Each shaft carries only one gear. Power is transmitted from driver to driven through intermediate gears.



17.2 2. Compound Gear Train

Definition

At least one shaft carries more than one gear. Used to obtain large velocity ratios in small space.

**Compound Gear Train**

17.3 3. Epicyclic Gear Train

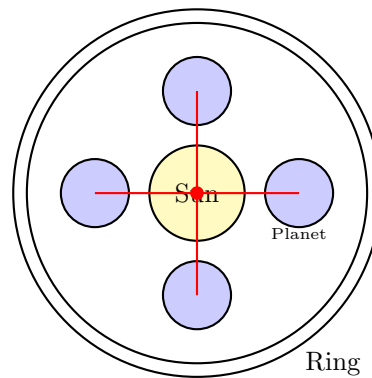
Definition

Gear train with at least one gear axis moving relative to the frame. Also called planetary gear train.

Components:

- **Sun gear:** Central gear

- **Planet gears:** Revolve around sun gear
- **Arm/Carrier:** Holds planet gears
- **Ring gear (Annulus):** Outer gear



Epicyclic Gear Train

18 Quick Reference Summary

Essential Formulas and Concepts

Kinematic Chain:

- $l = 2p - 4$ (links and pairs)
- $j = \frac{3}{2}(l - 1)$ (joints and links)

Degrees of Freedom: $n = 3(l - 1) - 2j - h$

Grashof's Law (Four Bar Chain): $L + S \leq P + Q$

Slider Crank Mechanism:

- Stroke Length: $L = 2r$
- Distance per minute: $= 4rN$
- $\sin \phi = \frac{\sin \theta}{n}$ where $n = l/r$

Quick Return Mechanism:

- Stroke Length: $L = \frac{2 \times AP \times CB}{AC}$
- Time Ratio: $= \frac{\beta}{\alpha}$

19 Classification Summary Tables

19.1 Types of Links

Type	Description
Rigid Link	No deformation, transmits both tension and compression
Flexible Link	Partly deformed, transmits only tension (belts, chains)
Fluid Link	Motion transmitted through fluid pressure

19.2 Types of Kinematic Pairs

Category	Type	Examples
5*Motion Type	Sliding	Piston-cylinder
	Turning	Shaft in bearing
	Rolling	Ball bearings
	Screw	Bolt-nut
	Spherical	Ball and socket
2*Contact Type	Lower Pair	Surface contact
	Higher Pair	Point/line contact
2*Closure Type	Self-closed	Mechanical connection
	Force-closed	External forces (spring)

19.3 Gear Classification

Shaft Position	Gear Type
Parallel	Spur, Helical, Double-helical
Intersecting	Straight bevel, Spiral bevel
Non-parallel, Non-intersecting	Skew bevel, Worm gear

Quick Recall Points

- **Machine vs Structure:** Machine has moving parts and transforms energy; Structure has no relative motion
- **Lower pair:** Surface contact — **Higher pair:** Point/line contact
- **Grashof's Law:** $L + S \leq P + Q$ for continuous motion
- **ODC to IDC:** Outer Dead Center to Inner Dead Center = one stroke
- **Quick return:** Cutting stroke angle > return stroke angle
- **Epicyclic:** At least one axis moves (planetary gears)

— End of Unit II Notes —